

Affordances, Signifiers and Constraints

Lecture 8

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Affordances, Signifiers and Constraints

Human-Computer Interaction (HCI)

THE POWER OF
EMAK-EMAK

SEIN KANAN BELOK KIRI

MASIH MENJADI
SEBUAH MISTERI

MP MERAH
PUTIH
.COM

PELESIRAN
DI NEGERI AING
WWW.PELESIRAN.AID



Bukan geng motor besar
yang jadi '**Raja Jalanan**'
tapi justru emak-emak
negara aing naik motor.



Dalam **Pasal 112 Undang-Undang Nomor 22 Tahun 2009** tentang Lalu Lintas dan Angkutan Jalanan, dijelaskan bahwa lampu sein merupakan hal yang wajib sebelum berbelok.

Sementara pengendara yang tak menyalakan lampu sein ketika berbelok akan dikenakan sanksi sebesar Rp250 ribu atau pidana: paling lama satu bulan.



JUST WHY ?





90's vs Millenium

Whose discoverability is better?



Coffeepot for Masochists

Fundamental Psychology of Interaction

Affordances

Signifiers

Constraints

Mapping

Feedback

Conceptual
Model

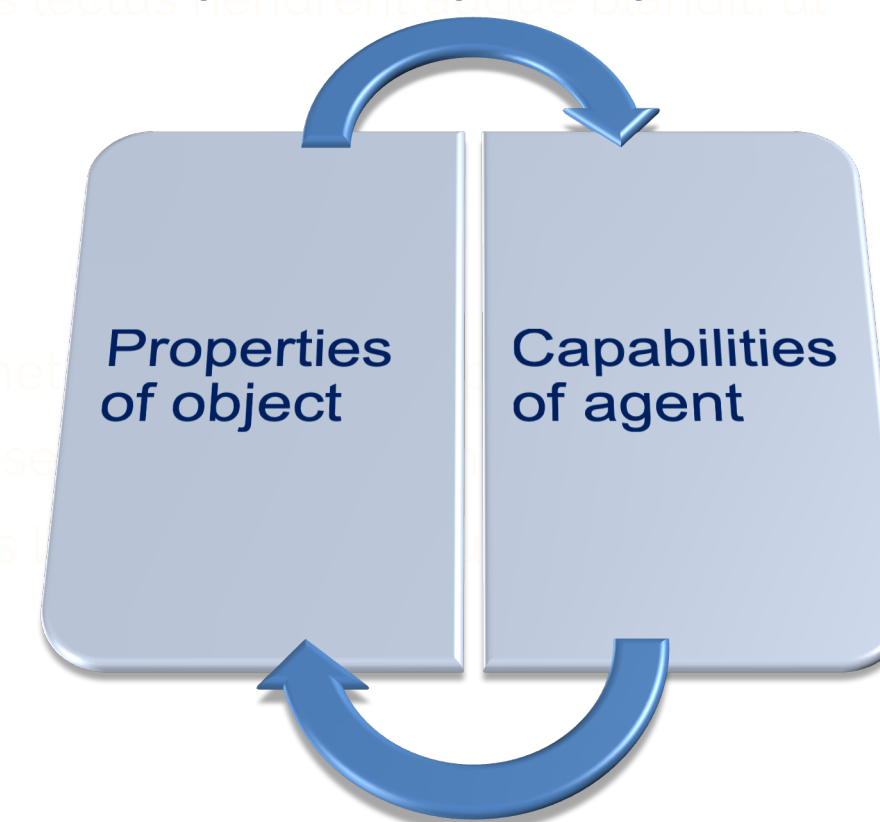
Affordance

Affordance



Relationship between a physical object and an interacting agent (human, animal, or robot).

Affordance determines the possibility in any ways the object can be used.





What actions are possible?

For example:

Door: We can open it!
PUSH/PULL the door?

How should the action take place?

For example:

The handle signifies where to PULL
The “PUSH” label on the plate signifies where to PUSH

Affordance vs. Signifier

Ambiguous Perceived Affordance

- This faucet in the picture demonstrates an ambiguous or confusing perceived affordance.
- Many people cannot directly tell which knob supplies hot or cold water, which one indicates the upper shower or lower faucet.



Affordances in Digital Interfaces

- In digital systems, affordances are **perceived rather than physical**.
- Examples in UI:
 - Raised buttons suggest clicking
 - Scroll bars suggest scrolling
 - Slider bars suggest dragging
 - Icons suggest actions
- Because users cannot physically touch digital objects, **visual design must communicate affordances**.

Real vs Perceived Affordances

- Two important concepts:
- **Real Affordance**
 - Actual action that can be performed.
 - Example:
A chair allows sitting.
- **Perceived Affordance**
 - What the user **believes** they can do.
 - Example:
A button **looks clickable**.

In UI design, **perceived affordances are critical**.

Example in Interface Design

- Examples of perceived affordances:
 - Buttons with **shadow or elevation**
 - Text fields with **outlined boxes**
 - Sliders with **handles**
- If an element **looks interactive**, users expect it to respond.
- If it looks static, users **will not try to interact**.

Signifier

Signifier

- A **signifier** is:

A signal that communicates where and how an action should take place

- Signifiers help users understand:
 - What actions are possible
 - Where actions should be performed
- How interaction works
- Signifiers **make affordances visible**.

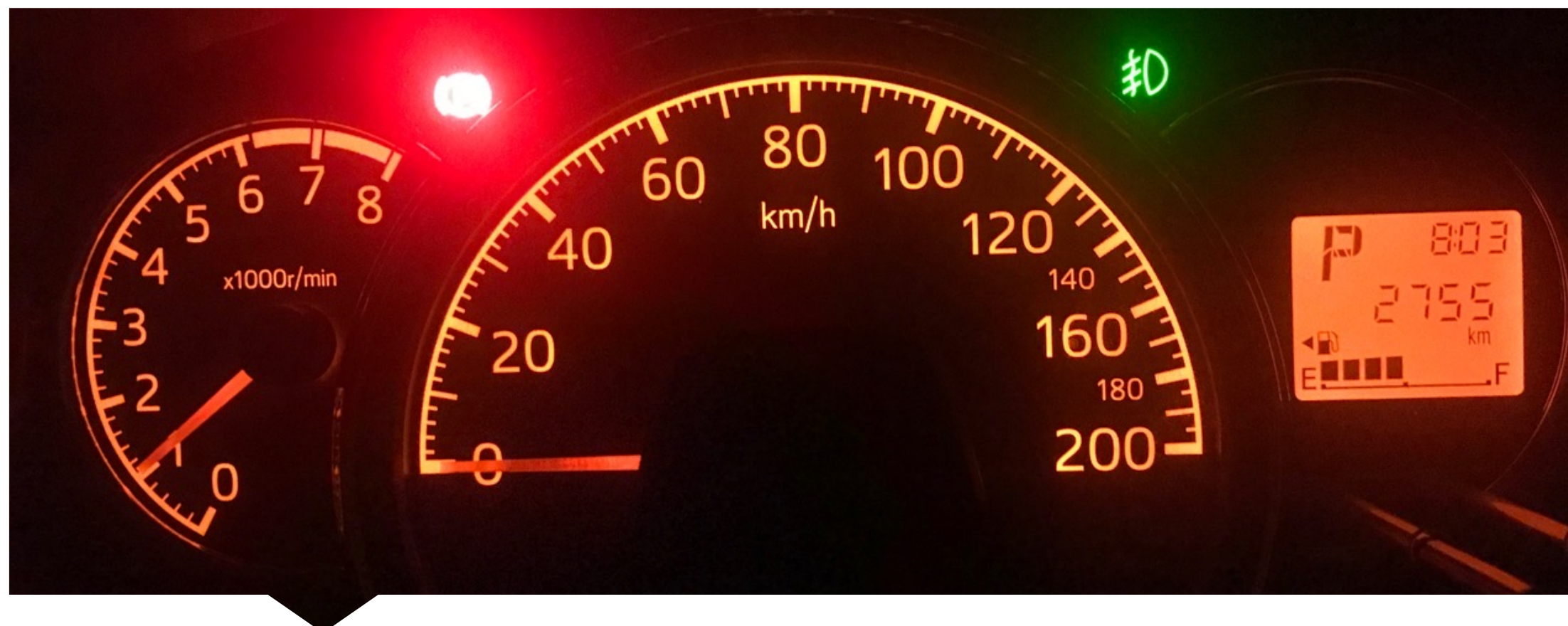
- Common examples:

- Labels
- Icons
- Arrows
- Instructions
- Highlighted buttons

- Example:

- A **door with a plate** indicates pushing.
- A **door with a handle** indicates pulling.





Clear Signifiers

The Graphic User Interface (GUI) on this dashboard is helpful to drivers because it helps them understand the affordances between them and the car they drive.

Signifiers in GUI

In GUI design, signifiers are more critical than affordances.

A signifier can be:

1. Word
2. Graphical illustration
3. Perceived affordance

Login

Signifiers in User Interfaces

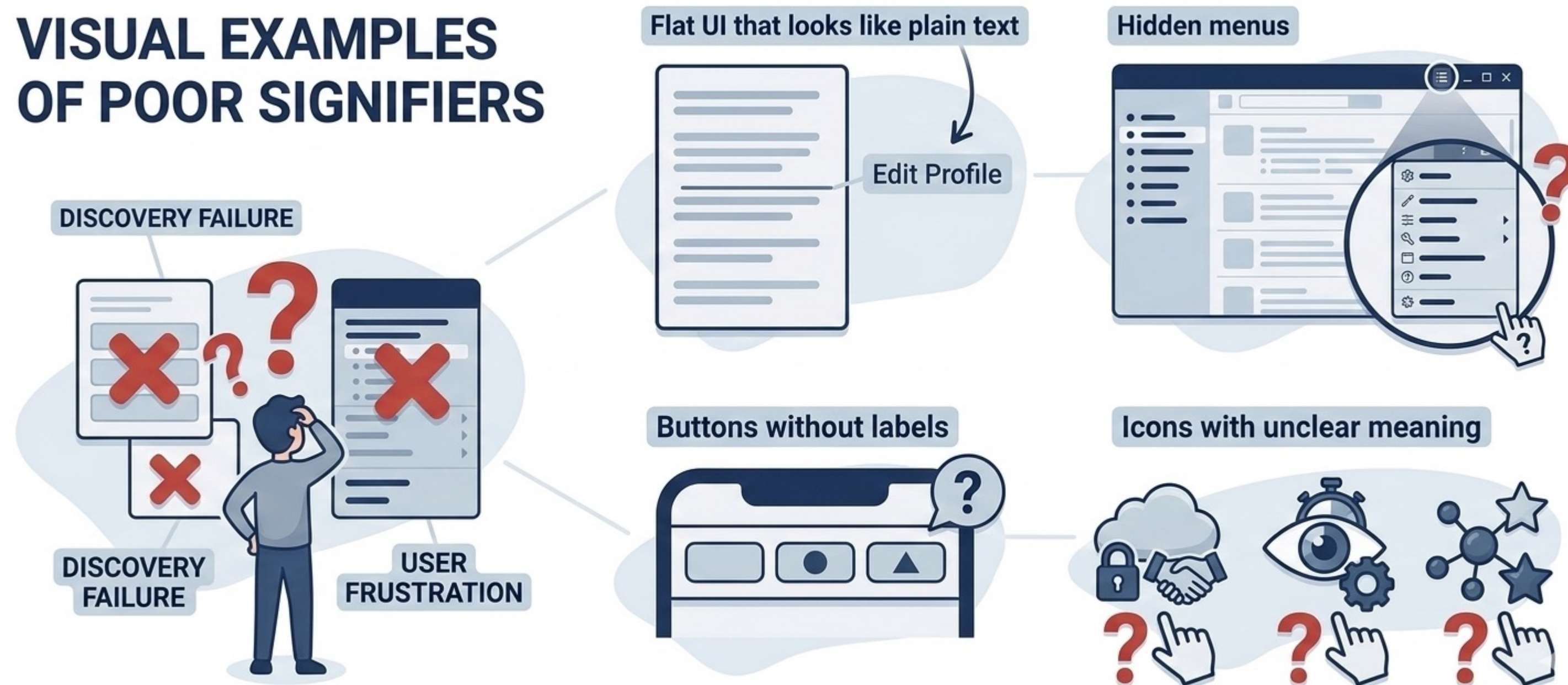
- Examples of UI signifiers:
 - Cursor change on hover
 - Highlight when hovering over buttons
 - Tooltips
 - Icons indicating function
 - Text labels
- Signifiers reduce **user confusion and uncertainty**.

Affordance vs Signifier

- Important distinction:
- Affordance:
 - What actions are possible
- Signifier:
 - How users know about those actions
- Example: Button
 - Affordance: clicking
 - Signifier: visual design indicating it is clickable

Poor Signifiers Example

VISUAL EXAMPLES OF POOR SIGNIFIERS



Constraint

What Is a Constraint?

- A **constraint** is:
 - A design element that restricts possible actions.
- Constraints guide users by **limiting incorrect actions**.
- They help:
 - Reduce errors
 - Simplify interaction
 - Improve usability

Why Constraints Are Important

- Without constraints, users can:
 - Perform incorrect actions
 - Enter invalid data
 - Break system workflows
- Constraints ensure users **only perform valid operations**

Types of Constraints

- Four common types:
 1. Physical constraints
 2. Logical constraints
 3. Cultural constraints
 4. Semantic constraints
- These help designers guide user behavior.

Physical Constraints

- Physical constraints restrict interaction based on **physical properties**.
- Examples:
 - USB plug orientation
 - Puzzle pieces
 - SIM card shape
- In UI:
 - Disabled buttons
 - Fixed input formats

Logical Constraints

- Logical constraints use **reasoning and relationships**.
- Example:
 - If only one slot remains in a puzzle, the remaining piece must fit there.
- In UI:
 - Step-by-step processes
 - Limited choices
- Example: Checkout process in e-commerce.

Cultural Constraints

- Cultural conventions influence interaction.
- Examples:
 - Red = stop
 - Green = go
 - Trash icon = delete
- These conventions depend on **shared cultural understanding**.

Semantic Constraints

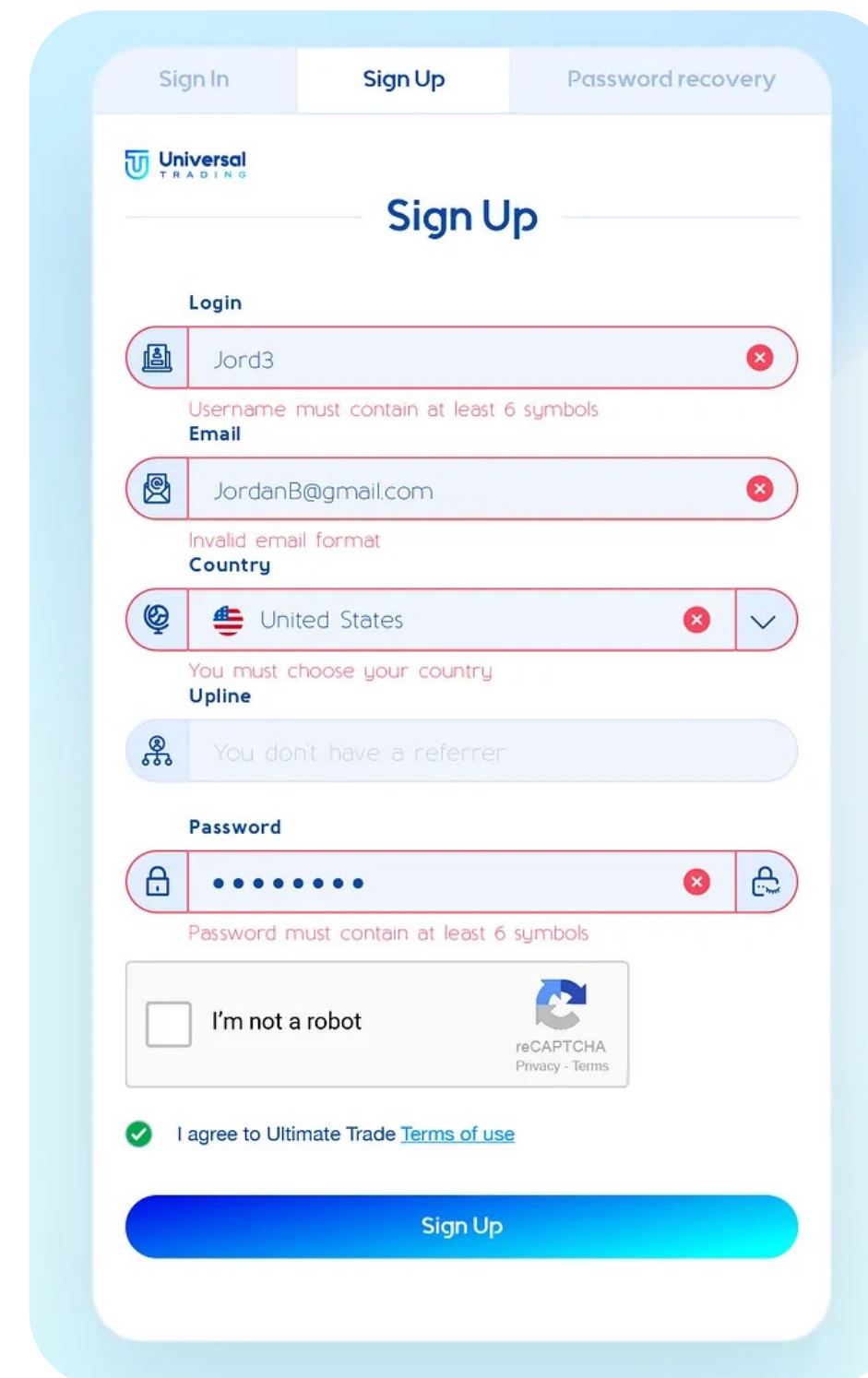
- Semantic constraints rely on **meaning and context**.
- Example:
 - A motorcycle rider faces forward.
 - Placing the rider backward would not make sense.
- In UI:
 - Icons and symbols must match their meaning.

Combining Affordances, Signifiers, and Constraints

- Good design integrates all three.
 - Affordances: Possible actions.
 - Signifiers: Signals explaining those actions.
 - Constraints: Restrictions preventing incorrect actions.
- Together they create **clear and usable interfaces**.

UI Design Example

- Example: Login Form
- Affordances:
 - Text field allows typing
- Signifiers:
 - Label "Username"
 - Cursor inside text field
- Constraints:
 - Password length rules
 - Required fields



The image shows a mobile app interface for a 'Sign Up' form. At the top, there are three tabs: 'Sign In', 'Sign Up' (selected), and 'Password recovery'. Below the tabs is the 'Universal TRADING' logo and the title 'Sign Up'. The form consists of several fields with error messages:

- Login:** A text field containing 'Jord3' with a red 'x' icon and the error message 'Username must contain at least 6 symbols'.
- Email:** A text field containing 'JordanB@gmail.com' with a red 'x' icon and the error message 'Invalid email format'.
- Country:** A dropdown menu showing 'United States' with a red 'x' icon and the error message 'You must choose your country'.
- Upline:** A text field containing 'You don't have a referrer'.
- Password:** A text field with masked characters (dots) and a red 'x' icon with the error message 'Password must contain at least 6 symbols'.

Below the password field is a checkbox labeled 'I'm not a robot' next to a reCAPTCHA logo. At the bottom, there is a green checkmark icon and the text 'I agree to Ultimate Trade [Terms of use](#)'. A large blue 'Sign Up' button is at the very bottom.

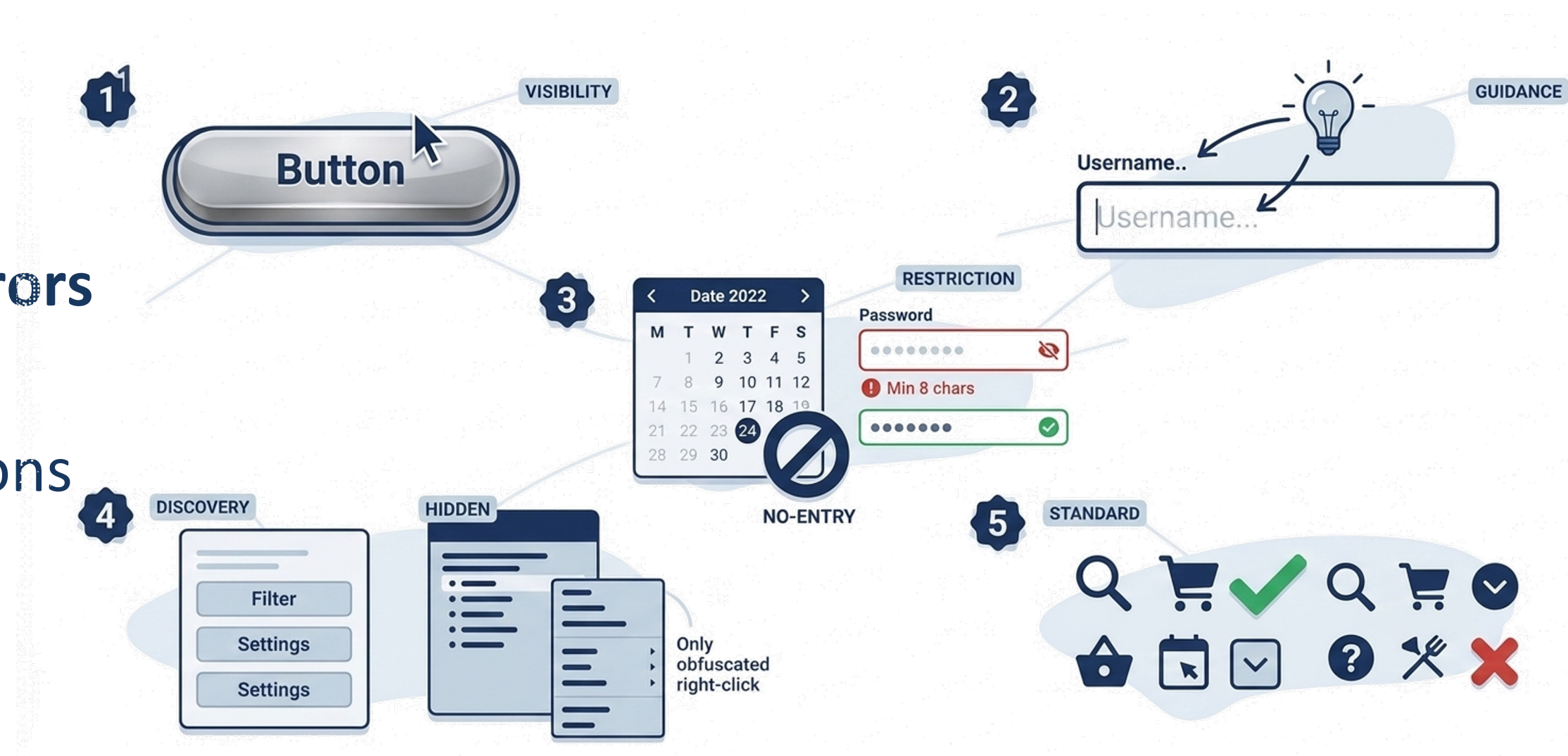


This is a reflection of the sign-up form shown above, appearing as if it's a duplicate or a second instance of the same form.

Design Guidelines

To design good interfaces:

1. Make affordances **clear**
2. Use signifiers **effectively**
3. Apply constraints **to prevent errors**
4. Avoid hidden functionality
5. Follow familiar design conventions



Assignment

- Tambahkan penjelasan konsep **Affordances**, **Signifiers**, dan **Constraints** pada tugas Perangkat Fisik berantarmuka buruk.
- Lakukan identifikasi terkait
 - Affordance pada antarmuka perangkat fisik
 - Signifiers yang memandu interaksi pengguna
 - Constraints yang mencegah kesalahan pengguna
- Tambahkan evaluasi dan saran dari antarmuka perangkat tersebut
- Pada analisis yang diberikan, jawablah pertanyaan berikut:
 - Apakah aksi yang mungkin dilakukan terdeskripsikan jelas bagi pengguna?
 - Apakah terdapat antarmuka yang ambigu dan membingungkan?
 - Bagaimana desain perlu diperbaiki untuk membuat interaksi lebih jelas dan intuitif?



Thank You

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